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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Machine Learning Engineer with 3 years of industry experience building and operating production **Python** ML and GenAI systems, including **self-hosted Llama-3 inference on a GPU EC2 instance** and the **AWS** infrastructure and **CI/CD** around it. On an MSc in Data Science (University of Bristol, awaiting result), trained **CNN, ResNet and ViT encoders with LSTM/attention decoders from scratch**, ran **multi-seed experiments** benchmarked against pre-trained VLMs, and co-built an evidence-backed scoring and ranking tool with **AstraZeneca** scientists, including its **data-preprocessing pipeline**. Used to working directly with domain experts to turn requirements into repeatable, working code; acted as **de facto team lead** for up to 4 interns. Seeking to help Mistral AI's researchers by building the ML pipelines and tooling that take work from experiment to production.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

GPU Inference, Serving & Cost

- Chose and operated a per-department model mix on quality and cost: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department's traffic, kept switchable to other models, with stronger **Azure OpenAI** outputs for decision makers; released to real users and changed the applications based on what was found.
- Designed the infrastructure for internal analytics platforms and GenAI applications on **AWS** (**EC2** including a **GPU** instance, **S3**, **ECR**, **Lambda**), packaged the **FastAPI** microservices in **Docker**, and built a one-click **Jenkins** and **GitLab CI/CD** deployment for separate dev and prod branches.

Machine Learning & Retrieval Systems

- Designed and built production **GenAI** applications end to end (**ReactJS** frontend, **Python FastAPI** microservices, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) serving Compliance, Client Service and Investment, with **Azure OpenAI embeddings** in a separate **pgvector** database per department.
- Built **tool-calling** agents that call **Python** functions to run **SQL**-backed fund comparisons and generate PDF reports on request; guardrailed behaviour with **system prompts**, versioned every prompt change in **Git** and had the Compliance department review each release.

Data Pipelines & Stakeholder Delivery

- Built a **Python** pipeline that parses over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches company names against the ones analysts track and sends a daily summary on a 9 AM **EventBridge** and **Lambda** schedule (run on **SageMaker**), saving **60 analyst hours a month**; parsed first with **AWS Textract**, then with **Azure OpenAI** after a rating agency began sending emails as images.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar** API covering every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with legacy analyst data; developed **Optimized SQL** queries, Functions and Procedures.
- Migrated financial analysts from Excel to **Power BI** dashboards, improving efficiency by **10 hours a week**; analysed customer data with **Python Pandas** and Market Basket Analysis (**Apriori** algorithm).

- Worked directly with stakeholders in Client Service, Compliance and Investment to turn requirements into working tools; acted as **de facto team lead** for up to **4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- University of Bristol MSc Data Science Dissertation in partnership with AstraZeneca: CellLineSelector (**Team project, co-built**. My focus: the **data-preprocessing pipeline** that reads 17 raw inputs and builds 14 clean, joined tables with identity resolution across sources, integrity checks and a build-manifest audit file; the **scoring and ranking** algorithm that ranks cancer cell lines for target inclusion and exclusion genes using per-gene scores across six evidence layers, calibrated across lineages; and the **RAG** assistant layer. Built with direct feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- University of Bristol MSc Applied Data Science Group Project: VisionCaptioner (**Team project**. Trained **SimpleCNN**, **ResNet-18** and a custom **ViT** encoder from scratch and compared symbolic, SBERT embedding-retrieval and **LSTM** decoders (global pooling or Bahdanau attention) generating image descriptions on three synthetic datasets (10,000 images each), across **five seeds** with Adam, evaluated with **SBERT cosine similarity, BLEU and ROUGE-L** and benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- Wingman: AI dating-coach web app with 108 LLM personas across 12 archetypes and vision-model scoring of uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with **Pydantic**, repaired when malformed and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- Project Hack 27: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- Encode Vibe Coding Hackathon: Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- University of Bristol Datathon sponsored by Lloyd's Bank: 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- Tools: Docker, Jenkins, GitLab (CI/CD), Git, Linux, AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker, Textract), Azure OpenAI, LlamaIndex, Nginx, Jupyter, Anaconda, VS Code, Claude Code.
- Languages: Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- Databases: PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite